

Shukriya! Ab main is outline ko **detailed** form mein likh raha hoon — aise jaise aap exam ki full tayyari ke liye chahtay hain. Neeche **pehle 5 points** ka **proper detailed version** diya gaya hai. Aap pehle is format ko dekh lijiye, agar pasand aaye to mein baqi points bhi isi style mein complete kar dunga.

📖 ICT Exam Detailed Notes (Part 1)

◆ 1. Difference between ICT and IT

✓ ICT (Information and Communication Technology)

- **Definition:** ICT involves all technologies used to handle telecommunications, broadcast media, audio-visual processing, intelligent building management systems, and network-based control and monitoring functions.
- **Scope:** Broader – Includes computers **and** communication technologies like mobile networks, internet, etc.
- **Examples:** Internet, email, social media, online learning, IoT devices, 5G networks.

✓ IT (Information Technology)

- **Definition:** IT refers specifically to the use of computers and software to manage information.
- **Scope:** Narrower – Focuses mainly on computing systems and software.
- **Examples:** Programming, data management, databases, cybersecurity, operating systems.

🔍 Comparison Table:

Aspect	ICT	IT
Full Form	Information & Communication Technology	Information Technology
Scope	Broad – includes IT + communication tools	Limited – mainly related to computing
Technologies Used	Internet, cloud, phones, communication software	Databases, programming, networking
Field	Education, media, health, government	Software companies, IT support

◆ 2. Digital Logic

✓ Logic Gates:

Logic gates are the building blocks of digital circuits. Each gate performs a specific logic operation using binary inputs (0 or 1).

Gate Name	Symbol	Operation	Output Rule
AND	$A \cdot B$	Multiplication	1 if both inputs are 1
OR	$A + B$	Addition	1 if any input is 1
NOT	$\neg A$ or A'	Inversion	Flips the input (0 becomes 1)
NAND	NOT + AND	Inverse of AND	0 only if both inputs are 1
NOR	NOT + OR	Inverse of OR	1 only if both inputs are 0

Gate Name	Symbol	Operation	Output Rule
XOR	Exclusive OR	1 if inputs are different	
XNOR	Exclusive NOR	1 if inputs are the same	

◆ 3. Truth Tables

A **truth table** shows all possible input values and the output for each logic gate.

✓ Example: AND Gate

A B A AND B

0 0 0

0 1 0

1 0 0

1 1 1

Truth tables help in understanding how logic circuits work for all input combinations.

◆ 4. Circuit Diagrams (NAND/NOR)

✓ What is a Circuit Diagram?

A graphical representation of a logic circuit using symbols of logic gates.

✓ NAND and NOR as Universal Gates:

- Any logic gate (AND, OR, NOT, etc.) can be created using only NAND or only NOR gates.
- This is useful for **cost-saving and efficient design** in electronics.

✓ Example:

Expression: $Y = A \text{ AND } B$

Can be drawn using:

- Direct AND gate, or
 - 2 NAND gates:
 - First NAND: $A \text{ NAND } B$
 - Second NAND: Output Nanded with itself (to invert it)
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◆ 5. Operating System – Definition

✓ What is an Operating System?

An Operating System (OS) is **system software** that manages hardware and software resources of a computer and provides services to programs and users.

✓ Main Functions:

1. **Memory Management** – Allocates memory to processes.
2. **Process Management** – Handles creation, scheduling, and termination of processes.
3. **Device Management** – Controls and coordinates input/output devices.
4. **File System Management** – Organizes data into files and directories.
5. **Security & Access Control** – Prevents unauthorized access.
6. **User Interface** – Command line or graphical interface.

✓ Types of Operating Systems:

- **Batch OS**
 - **Multitasking OS**
 - **Multiprogramming OS**
 - **Real-time OS**
 - **Distributed OS**
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Perfect! Yahan par mein aap ke handwritten outline ke **baqi tamam topics** ke liye detailed notes continue kar raha hoon — bilkul usi format mein jaisa aapko pasand aya.

📖 ICT Exam Detailed Notes (Part 2)

◆ 6. Multitasking

✓ Definition:

Multitasking is the capability of an operating system to execute **multiple tasks (processes)** simultaneously by quickly switching between them using **CPU scheduling**.

✓ Types:

- **Preemptive Multitasking:** OS decides which task runs for how long (e.g., Windows, Linux).
- **Cooperative Multitasking:** Running task voluntarily gives up control (e.g., older Mac systems).

✓ Benefits:

- Efficient use of CPU
 - Better user experience (multiple apps open)
 - Real-time task handling
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◆ 7. Multiprogramming

✓ Definition:

Multiprogramming is a method where **multiple programs are loaded into memory** and executed by the CPU one at a time to improve CPU utilization.

✓ **Example:**

While one program waits for I/O, the CPU executes another program.

✓ **Key Features:**

- Increases CPU efficiency
 - Needs memory management
 - Requires job scheduling
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◆ 8. Multiprocessing

✓ **Definition:**

Multiprocessing means a system has **two or more CPUs (processors)** working together to execute multiple processes simultaneously.

✓ **Types:**

- **Symmetric Multiprocessing (SMP):** All processors share memory and run tasks equally.
- **Asymmetric Multiprocessing (AMP):** One processor is master, others are slaves.

✓ **Advantages:**

- Faster processing
 - Better performance
 - Fault tolerance
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◆ 9. Registers (Flag, MAR, MBR, PC)

✓ **What are Registers?**

Registers are small, fast storage units in the CPU used to hold data temporarily during execution.

✓ **Types:**

- **Flag Register:** Stores status flags (Zero, Carry, Sign, etc.) that affect CPU decisions.
 - **MAR (Memory Address Register):** Holds memory addresses from which data will be fetched or written.
 - **MBR (Memory Buffer Register):** Temporarily stores data being transferred to/from memory.
 - **PC (Program Counter):** Contains the address of the next instruction to execute.
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◆ 10. Process Types

✓ Independent Process:

- Does **not depend on any other process** for execution.
- Runs in isolation.
- Example: A calculator app.

✓ Dependent Process:

- **Relies on other processes** (like parent-child processes).
- May share data or communicate with others.
- Example: A print job controlled by a parent process.

📖 ICT Exam Detailed Notes (Part 3)

◆ 11. Why and How OS Breaks Programs into Processes

✓ Why:

- To efficiently manage resources (CPU, memory).
- To isolate faults (one program crash doesn't crash whole system).
- For multitasking and process scheduling.

✓ How:

1. **Program is loaded** into memory.
2. OS creates a **Process Control Block (PCB)** for it.
3. Assigns process ID, memory space, registers.
4. Schedules the process using algorithms.

◆ 12. Difference Between SAD and OOAD

Feature	SAD (System Analysis & Design)	OOAD (Object-Oriented Analysis & Design)
Focus	Functions and data separately	Objects that combine data and behavior
Structure	Procedural and modular	Object-oriented (classes and objects)
Example	Flowcharts, DFDs	UML diagrams like Class & Use-Case
Reusability	Low	High (due to object reuse)

◆ 13. Assignment-Like Scenario for Team Structure or Process Models

✓ Scenario Example:

You are developing a university management system.

- **Team Structure:** Project Manager, Frontend Developer, Backend Developer, Tester.

- **Process Model:** Use **Waterfall** for fixed requirements OR **Agile** for flexible, fast-paced updates.
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◆ 14. Why Team Structure and Process Model are Important

- Ensure **clear roles and responsibilities**.
 - Improve **efficiency and collaboration**.
 - Define **how tasks will flow** (Agile, Waterfall, Spiral).
 - Reduce **errors and delays** in large projects.
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◆ 15. Difference in Product and Process

Feature	Product	Process
Definition	Final deliverable (software, app)	Method to create the product
Measurable	Quality, features, usability	Steps, tools, techniques
Focus	What is built	How it is built

◆ 16. Why It's Important to Launch a Product

- Timely launch creates **market advantage**.
 - Early feedback for **improvement**.
 - Helps establish **brand value**.
 - Delays can lead to **loss of customers** or **obsolete ideas**.
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◆ 17. Problem Statement

✔ What is a Problem Statement?

A clear, concise description of the issue that needs to be addressed by a solution.

✔ Format:

1. **Current situation**
2. **Problem description**
3. **Why it matters**
4. **Goals**

✔ Example:

“Students face difficulty managing class schedules manually. A web-based system is needed to automate timetable management and reduce errors.”

◆ 18. Use Case Diagram

✔ Definition:

A **use case diagram** shows how users (actors) interact with a system.

✔ Key Components:

- **Actors:** Users or systems
- **Use Cases:** Functionalities of the system
- **System Boundary:** The scope of the system

✔ Example Scenario:

Library Management System

- Actor: Student
 - Use Cases: Search Book, Borrow Book, Return Book
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◆ 19. Modes of Transmission / Components / Topologies

✔ Modes of Transmission:

- **Simplex:** One-way (e.g., TV broadcast)
- **Half-Duplex:** Two-way but one at a time (e.g., Walkie-Talkie)
- **Full-Duplex:** Two-way simultaneously (e.g., Telephone)

✔ Components of Data Communication:

1. **Message**
2. **Sender**
3. **Receiver**
4. **Transmission Medium**
5. **Protocol**

✔ Network Topologies:

- **Bus:** All devices share a single cable
 - **Star:** All devices connected to a central hub
 - **Ring:** Each device connected to two others
 - **Mesh:** Devices are interconnected
 - **Hybrid:** Mix of topologies
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◆ 20. Definitions from Presentations

Revise all bold or highlighted terms from your class slides such as:

- Data
- Information
- Bandwidth
- Latency

- IP Address
 - Compiler vs Interpreter
 - Algorithm, Flowchart
 - Cloud Computing, IoT, etc.
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